

POSITION-SPECIFIC ACCELERATION AND DECELERATION PROFILES IN ELITE YOUTH AND SENIOR SOCCER PLAYERS

JEPPE F. VIGH-LARSEN, ULRIK DALGAS, AND THOMAS B. ANDERSEN

Section for Sport Science, Department of Public Health, Aarhus University, Aarhus, Denmark

ABSTRACT

Vigh-Larsen, JF, Dalgas, U, and Andersen, TB. Position-specific acceleration and deceleration profiles in elite youth and senior soccer players. *J Strength Cond Res* 32(4): 1114–1122, 2018—The purpose of the study was to characterize and compare the position-specific activity profiles of young and senior elite soccer players with special emphasis put on accelerations and decelerations. Eight professional senior matches were tracked using the ZXY tracking system and analyzed for the number of accelerations and decelerations and running distances within different speed zones. Likewise, 4 U19 and 5 U17 matches were analyzed for comparison between youth and senior players. In senior players, the total distance (TD) was $10,776 \pm 107$ m with 668 ± 28 and 143 ± 10 m being high-intensity running (HIR) and sprinting, respectively. Number of accelerations and decelerations were 81 ± 2 and 84 ± 3 , respectively, with central defenders performing the lowest and wide players the highest number. Declines were found between first and second halves for accelerations and decelerations ($11 \pm 3\%$), HIR ($6 \pm 4\%$), and TD ($5 \pm 1\%$), whereas sprinting distance did not differ. U19 players performed a higher number of accelerations, decelerations, and TD compared with senior players. In conclusion, differences in the number and distribution of accelerations and decelerations appeared between player positions, which is of importance when monitoring training and match loads and when prescribing specific training exercises. Furthermore, youth players performed as much high-intensity activities as senior players, indicating that this is not a discriminating physiological parameter between these players.

KEY WORDS accelerations, decelerations, fatigue, monitoring

Address correspondence to Jeppe F. Vigh-Larsen, jeppefaged@hotmail.com.

32(4)/1114–1122

Journal of Strength and Conditioning Research
© 2017 National Strength and Conditioning Association

1114 ^{the}Journal of Strength and Conditioning Research

INTRODUCTION

Previous studies have shown that competitive soccer games are intermittent in nature with approximately 1,000–1,500 activities performed corresponding to a change in activity every 4–6 seconds (3). The majority of activities are performed at low intensity but interspersed by brief intense actions such as short high-intensity running bouts, tackles, or forceful accelerations and decelerations resulting in an average heart rate of approximately 85% heart rate max corresponding to approximately 75% $\dot{V}O_2$ max (3,12,25,33). This means that the aerobic system is the primary fuel source throughout a match, whereas at the same time, the anaerobic system is highly active during intense periods, as evidenced by measures of blood lactate and creatine phosphate levels (4). Research has used video-based or computer-assisted time-motion analysis, reporting the distances covered in different categories during soccer matches based on speed thresholds. Accordingly, consensus exists that soccer players on average cover 9–13 km depending on playing position, tactical formation of the team, and strength of the opponent team but relatively independent of performance level (16,17,25,30,33). Likewise, approximately 600–900 m and approximately 200–300 m are covered in high-intensity running and sprinting, respectively, but with some discrepancy existing in the literature (9,10,12,15,17,23,25,32,36). Accordingly, high-intensity running and sprinting discriminate high- and low-level players and differ between different periods throughout a match (12,25). This line of research has provided useful insight into position-specific game demands and provided coaches and sport scientists with information required to develop specific training programs, monitor player load, and individualize recovery strategies. However, using distance and speed measures only may underestimate the workload on certain players, as recent studies have shown that several high-intensity efforts may occur when accelerating and decelerating, despite moving at low speed (20,28,36). For example, one study (36) found that accelerations and decelerations comprised 12–17% of the total player load. Likewise, a 6–8% difference in total load estimation has been reported when comparing speed- vs. acceleration-derived monitoring techniques (20,28).

Furthermore, accelerations and decelerations require high rates of force development and greater neural activation of the working muscles compared with constant speed running, and they contribute significantly to muscle damage possibly caused by the substantial eccentric muscle contractions during these activities (1,18,26). Therefore, quantification of accelerations and decelerations are important to achieve a deeper understanding of position-specific game demands and provide more detailed feedback, allowing improved individualization of training prescription and recovery. In addition, quantification of accelerations and decelerations could prove to be valid performance markers, as recent studies have suggested that these activities are more sensitive to fatigue than running distance at different speeds. Moreover, a recent study (1) observed that the number of accelerations and decelerations declined more during a match than high-intensity running or sprinting. Accordingly, another study (2) found that the frequency of accelerations and decelerations decreased markedly during an intensive youth tournament but with no concomitant changes in other running related outcomes. Furthermore, the number of accelerations and decelerations during games may also be a discriminating factor between youth and senior players, as has previously been shown for high-intensity running and sprinting in high- vs. moderate-level players (25). Thus, determining the sensitivity and validity of accelerations and decelerations as critical performance markers is important when monitoring fatigue development during games to optimize substitution and player rotation strategies during congested fixtures and when determining the game demands in youth vs. senior players (14). Consequently, more information is needed on position-specific game demands, differences between youth and senior players, and the validity of accelerations and decelerations as markers of physical match performance. Thus, the aim of the present study were (a) to characterize the position-specific activity profiles of senior elite soccer players with a special focus put on accelerations and decelerations and (b) to compare the activities of senior players with those of young elite players. It was hypothesized that substantial differences would exist between positional demands in terms of the number of accelerations and deceleration and that these activities would be sensitive to fatigue during games. Furthermore, we expect a superior number of accelerations and decelerations in elite senior vs. youth matches.

METHODS

Experimental Approach to the Problem

The activity profile of 14 outfield soccer players competing for a single club from the best Danish soccer league (The Danish Superliga) during one half season (spring 2015) was monitored. The data set included all home matches ($n = 8$ matches, 54 observations), and only players completing the full duration of matches were included in the analysis. Players were separated according to position, the sample

including 3 different central defenders ($n = 12$ observations), 3 wide defenders ($n = 12$ observations), 3 central midfielders ($n = 16$ observations), 3 wide midfielders ($n = 9$ observations), and 2 attackers ($n = 5$ observations). Furthermore, players were grouped in wide players (wide defenders and wide midfielders, $n = 21$ observations) and central players (central defenders, central midfielders, and attackers, $n = 32$ observations) to analyze whether specific match activities were performed by players in the lateral vs. central areas of the pitch, as previously suggested in a recent study (23). Data were analyzed as total match activities to describe the overall position-specific game demands, and a comparison between first vs. second half was performed to detect fatigue during the game. Finally, match activities were divided in 15-minute periods and then compared with the purpose of evaluating fatigue development in more detail during different phases during a game, as has previously been done in team sports (25). During the season, the team ended up as national champions with a total of 7 wins and 1 defeat (score line 21–6) in the recorded matches. Besides data from the senior team, league matches from the same period for the U19 ($n = 4$ matches, 25 observations) and U17 ($n = 5$ matches, 34 observations) youth teams from the same club playing in the best division in their respective age groups were analyzed for a comparison between youth and senior players. The sample included 7, 6, 8, 7, and 7 observations for U17 and 7, 7, 3, 3, and 5 observations for U19 for central defenders, wide defenders, wide midfielders, attackers, and central midfielders, respectively. However, youth and senior matches were only compared in terms of overall game demands with no analysis of positional differences between the levels of play.

Subjects

Elite male soccer players ($n = 14$) from 1 single team playing in the best Danish division participated in the study (age, 26.5 ± 1 years; range, 21–31 years; height, 184 ± 2 cm; body mass, 82 ± 2 kg). Likewise, elite U19 ($n = 13$) (age, 18.5 ± 0.2 years; range, 17–20 years) and U17 ($n = 17$) (age, 16.3 ± 0.2 years; range, 15–17 years) youth players from the same club playing at the highest level in their age groups participated, but anthropometric data were not available for these players. Both youth and senior players had undergone an 8-week preseason leading into the season following a 4-week midseason winter break and were training 6–7 times a week (5–6 on-field session and 1 strength session). No specific guidelines were given during the study period because all subjects followed their habitual training, nutrition, and hydration strategies. The study was performed in accordance with the Declaration of Helsinki in guidance with the local ethics committee (De Videnskabetiske Komiteer for Region Midtjylland), and each subject or parent (if the subject was younger than 18 years old) gave his or her informed written consent to participate in the study. There was no specific application to the De Videnskabetiske Komiteer for Region Midtjylland Ethics Committee for the present study, as there exists a general permission given to the Department,

TABLE 1. Locomotor activities during games for each position, groups of players, and team average.*

	Attackers	Central midfielders	Wide midfielders	Wide defenders	Central defenders	Wide players (WM + WD)	Central players (CM + CD + A)	Team average
Accelerations (n)								
First half	42 ± 4	41 ± 2	44 ± 3	43 ± 3	36 ± 3	44 ± 2	39 ± 2	41 ± 1
Second half	35 ± 4	35 ± 2	40 ± 3	44 ± 3	29 ± 3	42 ± 2	33 ± 2†	37 ± 1†
Full match	80 ± 6	81 ± 5	89 ± 5‡	91 ± 4‡	68 ± 4	90 ± 3§	76 ± 3	81 ± 2
Decelerations (n)								
First half	46 ± 4	43 ± 2	45 ± 3	48 ± 3	32 ± 3	47 ± 2	39 ± 2	43 ± 1
Second half	38 ± 4	37 ± 2	42 ± 3	47 ± 3	25 ± 3	45 ± 2	33 ± 2†	38 ± 1†
Full match	87 ± 8‡	83 ± 4‡	91 ± 7‡	99 ± 4 ‡	61 ± 3	96 ± 4§	75 ± 3	84 ± 3
Accelerations + decelerations (n)								
First half	88 ± 7	84 ± 4	89 ± 5	91 ± 5	68 ± 5	90 ± 3.7	79 ± 3	84 ± 2
Second half	73 ± 7	72 ± 4†	82 ± 5	91 ± 5	54 ± 5†	87 ± 3.7	66 ± 3†	74 ± 2†
Full match	167 ± 13	164 ± 7‡	180 ± 12‡	190 ± 8‡	129 ± 6	186 ± 7§	151 ± 5	165 ± 5
High-intensity running (m)								
First half	385 ± 40	403 ± 22	337 ± 30	355 ± 26	212 ± 3	347 ± 23	331 ± 19	338 ± 13
Second half	286 ± 40	317 ± 22†	352 ± 30	375 ± 26	175 ± 3	365 ± 23	260 ± 19†	301 ± 13†
Full match	693 ± 59‡	763 ± 43‡	724 ± 59‡	766 ± 38‡	398 ± 32	748 ± 33§	615 ± 39	668 ± 28
Sprinting (m)								
First half	74 ± 18	97 ± 10	86 ± 14	64 ± 12	40 ± 12	73 ± 9	73 ± 8	72 ± 6
Second half	57 ± 18	50 ± 10†	90 ± 14	85 ± 12	35 ± 12	87 ± 9	46 ± 8†	63 ± 6
Full match	143 ± 34	160 ± 20‡	185 ± 23‡	156 ± 17‡	77 ± 17	168 ± 14§	127 ± 13	143 ± 10
Total distance (m)								
First half	5,160 ± 142	5,643 ± 79	5,000 ± 106	5,284 ± 92	4,860 ± 92	5,162 ± 93	5,285 ± 74	5,190 ± 47
Second half	4,802 ± 142	5,254 ± 79†	4,777 ± 106	5,086 ± 92	4,541 ± 92†	4,953 ± 93	4,926 ± 74†	4,892 ± 47†
Full match	10,551 ± 217	11,565 ± 124¶#**‡	10,352 ± 162	10,996 ± 138#‡	9,933 ± 140	10,754 ± 130	10,789 ± 157	10,776 ± 107

*Data are presented as mean ± SEM; first half = first 45 minutes; second half = second 45 minutes; full match = 90 minutes + additional time. Comparisons are made of full match data for each parameter between positions (attackers vs. central midfielders vs. wide midfielders vs. wide defenders vs. central defenders) and between groups (wide players vs. central players) and between first half vs. second halves within each position and each group.

†Less than first half ($p \leq 0.05$).

‡Higher than central defenders.

§Higher than central players.

||Higher than central midfielders.

¶Higher than attackers.

#Higher than wide midfielders.

**Higher than wide defenders.

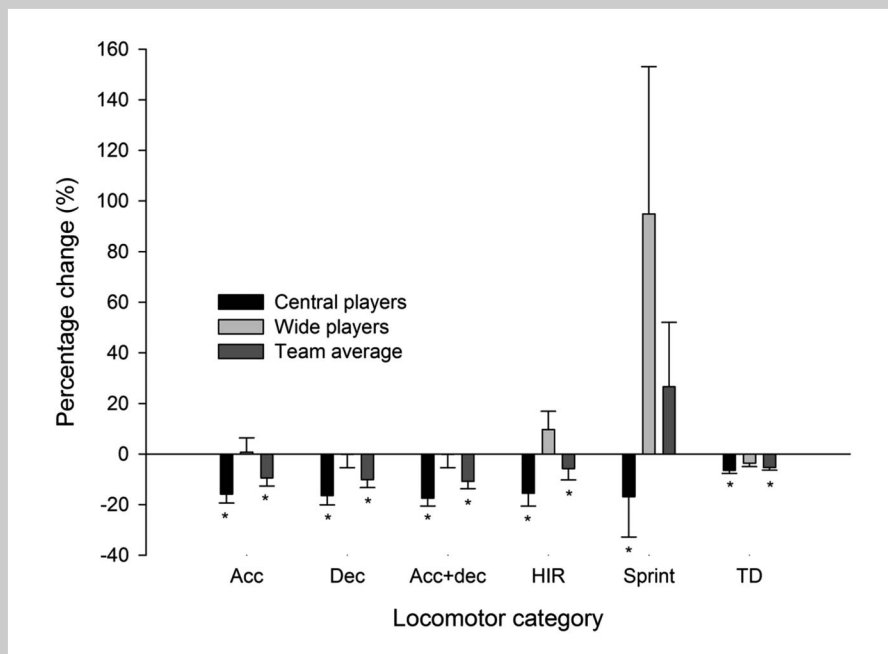


Figure 1. First vs. second half percentage change in each locomotor category for central players, wide players, and team average. Acc = accelerations; Dec = decelerations; Acc + dec = accelerations and decelerations; HIR = high-intensity running; TD = total distance. Data are presented as mean percentage change $\pm$ SEM. *Significant change ($p \leq 0.05$).

to perform noninvasive movement analyses, using the Committee's guidelines on informed consent and the like.

Procedures

The movements of all players were tracked using a local positioning system, the ZXY tracking system (<http://chyronhego.com/sports-data/zxy>), using small body worn sensors positioned in a belt around the waist of each player. The belt contains electronic sensors that register effort-sensor signals, axial rotation speeds, and body movement on heading in 3 directional axes as described by Bendiksen et al. (6). These data are then transmitted simultaneously to a stationary sensor system mounted around the tribune roof based on 2.45 GHZ ISM bands used for radio communication. The stationary system then computes the positional data from each belt in the field by providing advanced vector-based processing of the radio signals received with a sampling rate of 20 Hz (6). These data are then stored in a SQL server database for subsequent analysis. The reliability of the system has recently been described by Ingebrigtsen et al. (23), presenting a test-retest reliability, assessed by the intraclass correlation coefficient of $r = 1.0$, $r = 0.999$, and $r = 0.999$ ($p = 0.000$) for the x and y positions and total distance (TD) covered, respectively. In addition, they reported no significant differences between the test-retest data for all variables using a Student's paired t -test or Bland-Altman plots

for the x and y position or TD. These findings indicate a good reliability of the ZXY tracking system.

The ZXY tracking systems provided information regarding distance within different speed zones and the number of acceleration and decelerations performed each minute. The following locomotor categories were selected for analysis of the specific running activities of interest in the present study: High-intensity running (from 19.8 to 25.2 $\text{km} \cdot \text{h}^{-1}$) and sprinting ($\geq 25.2 \text{ km} \cdot \text{h}^{-1}$) measured as distance covered within this speed threshold. Furthermore, accelerations and decelerations were measured as number of counts as defined by 4 event markers: (a) The acceleration had to reach a minimum of 1 $\text{m} \cdot \text{s}^{-2}$ to mark the start of the acceleration event. (b) The acceleration

had to reach and exceed the acceleration limit (2 $\text{m} \cdot \text{s}^{-2}$) to be counted as an acceleration, and (c) the acceleration had to remain above this limit for at least half a second. (d) Finally, the acceleration event ended when the acceleration went below the minimum acceleration speed. For decelerations, the same events applied, just with negative signs. In the analysis, accelerations and decelerations were also summarized and evaluated collectively to obtain a single expression of the acceleration and deceleration profile of the players.

Statistical Analyses

Data are presented as mean $\pm$ SEM. A Shapiro-Wilk test was used to check data normality. To test the primary hypotheses, a 2 way (time $\times$ group) for repeated-measures analysis of variance (ANOVA) was used to identify differences between first and second halves and playing position for each dependent variable of interest, whereas differences between 15-minute periods and playing positions overall were analyzed using a 1-way ANOVA for repeated measures. Likewise, to test the secondary hypothesis, a 1-way ANOVA for repeated measures were used to analyze differences between senior, U19, and U17 matches. In the comparison between youth and senior matches, data were expressed as distance or counts per minute to compare the match activities despite different match durations. When a significant interaction was found, multiple comparison procedures were applied using the

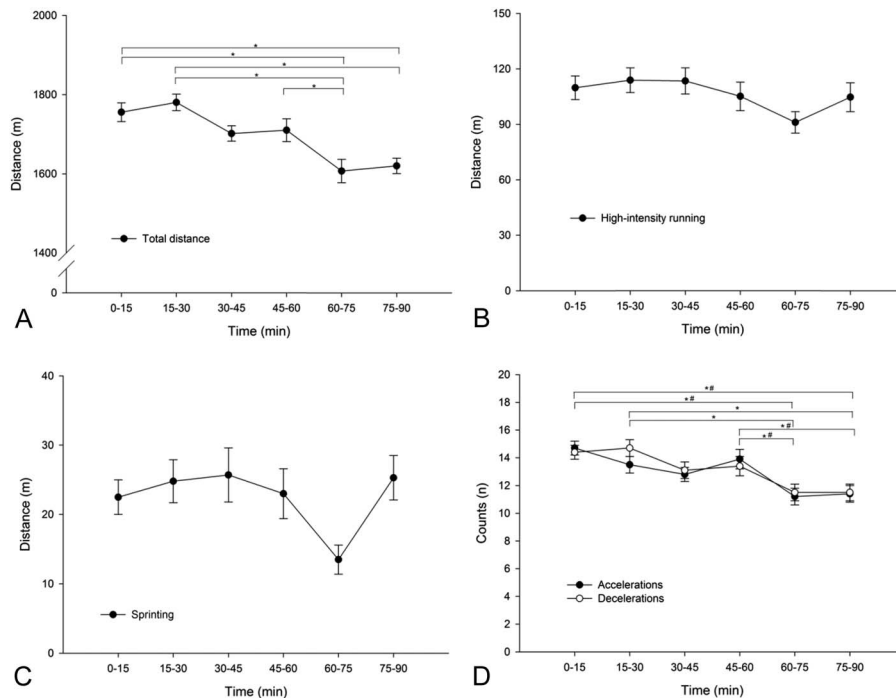


Figure 2. Total distance in 15-minute periods (A), high-intensity running in 15-minute periods (B), sprint distance in 15-minute periods (C), and acceleration and decelerations in 15-minute periods (D). Data are presented as mean $\pm$ SEM. *Significant difference ($P \leq 0.05$). In (A), * indicates significant difference for decelerations ($p \leq 0.05$) and # indicates significant difference for accelerations ($p \leq 0.05$).

Holm-Sidak method to detect where the significant difference existed. Significance was accepted at $p \leq 0.05$. SigmaPlot was used for all analyses (SigmaPlot for Windows version 11.0; Systat Software Inc., London, United Kingdom).

RESULTS

Players covered on average $10,776 \pm 107$ m TD during a full match, with 668 ± 28 m and 143 ± 10 m being high-intensity running and sprinting, respectively. Central midfielders covered significantly more TD than all other positions ($p \leq 0.05$), whereas wide defenders covered significantly more TD than central defenders ($p \leq 0.05$) and wide midfielders ($p \leq 0.05$), with no other significant differences between positions (Table 1). In terms of high-intensity running, all positions covered significantly more distance than the central defenders ($p \leq 0.05$) with no other significant differences being observed between other positions (Table 1). Regarding sprinting distance, the same picture was evident with all positions, except attackers, covering significantly more sprinting than the central defenders ($p \leq 0.05$). In general, when comparing wide players (wide defenders and wide midfielders) with central players (central defenders, central midfielders, and attackers), a significantly higher sprinting and high-intensity running distance were observed for the wide players ($p \leq 0.05$).

The total number of accelerations and decelerations for all the positions were on average 165 ± 5 , whereas the number of accelerations and decelerations in isolation were 81 ± 2 and 84 ± 3 , respectively (Table 1). When comparing the different playing positions, all positions, except attackers, had significantly more accelerations and decelerations than the central defenders ($p \leq 0.05$), whereas wide defenders had significantly more decelerations than central midfielders as well ($p \leq 0.05$). All positions had more decelerations than central defenders; however, no differences existed regarding the number of accelerations between attackers, central midfielders, and central defenders because only wide defenders and wide midfielders had a higher number of these actions than central defenders ($p \leq 0.05$). When grouping the players in wide and central players, wide players had a significantly higher amount of accelerations and decelerations than central players ($p \leq 0.05$).

When comparing the first and second halves, there was a significant decline in the amount of high-intensity running distance (first half vs. second half: 338 ± 13 vs. 300 ± 13 m; $p \leq 0.05$), acceleration + deceleration (first half vs. second half: 84 ± 3 vs. 74 ± 2 ; $p \leq 0.05$), accelerations (first half vs. second half: 41 ± 1 vs. 37 ± 1 ; $p \leq 0.05$), decelerations (first half vs. second half: 43 ± 1 vs. 38 ± 1 ; $p \leq 0.05$), and TD covered (first half vs. second half:

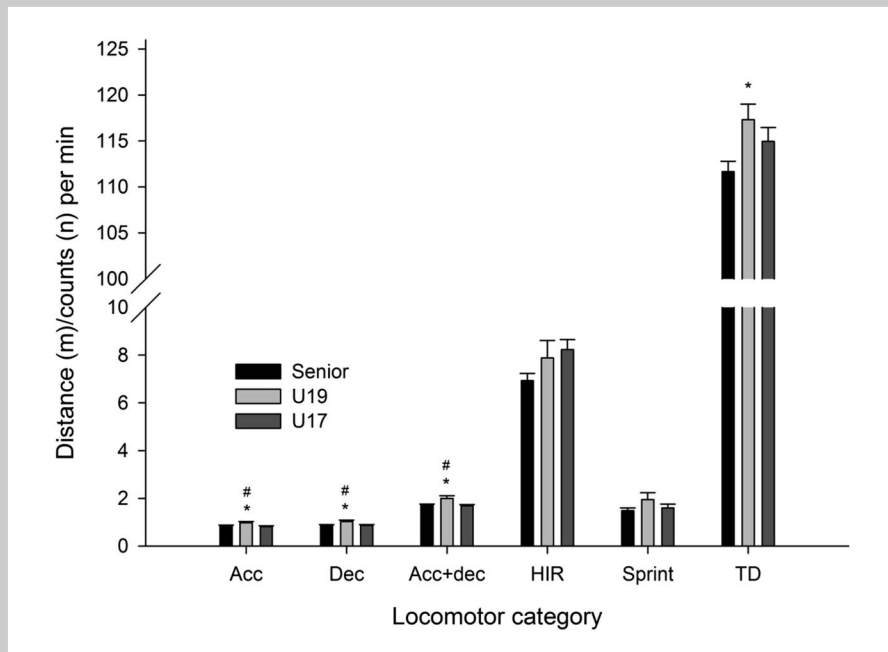


Figure 3. Differences between senior, U19, and U17 for all locomotor categories expressed as distance (high-intensity running, sprint, total distance) or counts (accelerations and decelerations) per minute. Data are presented as mean $\pm$ SEM. Acc = accelerations; Dec = decelerations; Acc + dec = accelerations and decelerations; HIR = high-intensity running; TD = total distance. *Significantly higher than senior; #Significantly higher than U17 ($p \leq 0.05$).

5,190 $\pm$ 47 vs. 4,892 $\pm$ 47 m; $p \leq 0.05$) but no significant change in the amount of sprinting (first half vs. second half: 72 $\pm$ 6 m vs. 63 $\pm$ 6 m). For individual positions, several differences existed between halves (Table 1), and when grouping players in wide and central players, the central players had a significant decrease in every performance variable in contrast to wide players where no differences between halves existed ($p \leq 0.05$) (Figure 1).

Looking into the match divided in six 15-minute periods (extra time excluded), a significant overall decline was found in the number of accelerations and decelerations between the first two 15-minute periods of the match and the last two 15-minute periods, as well as between the first 15-minute period of the second half and the last two 15-minute periods ($p \leq 0.05$) (Figure 2). Examining the same periods for high-intensity running or sprinting, no significant differences were found between any 15-minute periods. For TD, however, a similar picture was observed as for accelerations and decelerations, as more distance was covered during the first two 15-minute periods of the match compared with the last two 15-minute periods, as well as a difference existing between the first 15-minute period of the second half and the following 15-minute period ($p \leq 0.05$).

When comparing the number of accelerations and decelerations or distance covered per minute for senior, U19, and U17, a significant difference was found for TD, with players

covering more TD during the U19 compared with the senior matches but not the U17 matches ($p \leq 0.05$) (Figure 3). Also, a higher number of accelerations and decelerations were found during the U19 matches when compared with both the senior and the U17 matches ($p \leq 0.05$). However, no differences were found for distance covered during high-intensity running or sprinting.

DISCUSSION

The main findings of the present study were that positional differences existed for the number of accelerations and decelerations with central defenders performing the lowest number of all positions and wide players in general performing more than central players as hypothesized. Playing positions were also related to the change in work rate between halves because only central players had a significant decline in

performance during the second half. Contrasting the hypotheses, the senior players did not have higher game demands compared with the youth players because these players performed as much high-intensity activities per minute and in some instances even more accelerations and decelerations during the U19 matches. Finally, accelerations and decelerations seems to be sensitive and valid markers of physical match performance, as changes were captured during the matches with significant differences existing between halves and between separate 15-minute periods.

The total number of accelerations and decelerations in the present study were 165 (accelerations vs. decelerations: 81 vs. 84), which is comparable to what was found in a study on Norwegian elite players where 91 accelerations were recorded (23). In contrast, less accelerations and decelerations were found in another study on Norwegian elite players where 76 and 54 accelerations and decelerations were recorded, respectively (36). Likewise, 65 accelerations were captured in a study on Australian elite players (38), whereas English Premier league players were measured to decelerate only 54 times during games (7). However, the first study used a higher threshold for defining accelerations (>2.78 vs. >2 m·s⁻²), whereas the latter was based on subjective observations of video material and thus not directly comparable to our findings. Contrary to this, other studies have

reported a higher number of accelerations and decelerations than in the present study. For example, top-class players accelerated 119 times during a game (10), whereas Australian A-League players were reported to accelerate and decelerate 113 and 148 times, respectively (40). Some of these discrepancies can be related to the use of different thresholds for defining accelerations and decelerations and the use of different tracking systems. Accordingly, data obtained by different sports tracking systems, different GPS units, or analyzed with different software should be compared with caution as measurement differences can occur, especially when systems with different sampling rates are used to track rapid movements such as accelerations and decelerations (13,32,39). In the present study, the ZXY tracking system was used as in the studies on Norwegian players (23,36), whereas different systems were used in the other studies, thus limiting direct comparisons of data.

Across playing positions, wide players accelerated and decelerated more often than central players in the present study, and central defenders less than all other positions. This is consistent with previous findings (23,36,38), except that no differences were observed between central defenders and any other positions in the latter study. In contrast, one study found defenders to accelerate and decelerate more than attackers (40); however, no distinction was made between wide and central defenders, which could explain these findings. Thus, an overall agreement exist that players in wide positions have a higher number of accelerations and decelerations than players in central positions and that central defenders perform the lowest number despite some conflicting evidence. Related to this, earlier studies have shown that the metabolic power output of accelerations are elevated in comparison to running at constant speed and that these activities are correlated with the degree of post match fatigue. This means that particularly the physical demands for wide players appear to be even higher and for central defenders even lower than when just interpreting measures of speed and distance (1,20,28,36).

The TD covered in the present study was $10,776 \pm 107$ m, which is in line with the distances reported in the literature at different levels of play (10,16,25). Total distance covered in high-intensity running and sprinting was 668 ± 28 m and 143 ± 10 m, respectively, which is less than reported in previous studies on high-level players (high-intensity running: ~ 600 – 900 m; sprinting: ~ 200 – 300 m), despite using the same velocity threshold (>25.2 km·h⁻¹) (9,10,12,15,17,23,36). The differences may reflect the intensity of play in the Danish league and could be influenced by fitness levels, style of play, level of the opposition, or the use of different tracking systems, which has previously shown to affect the distances recorded, especially concerning high-intense activities (13,32).

Overall significant decreases between halves were observed for all parameters, except for sprinting distance. The amount of accelerations and decelerations decreased by

$11 \pm 3\%$, high-intensity running by $6 \pm 4\%$, and TD covered by $5 \pm 1\%$. This is partly in line with the literature (11,16,19,23,34,36) because decreases of 0–5% have typically been observed for TD, whereas sprinting distance have shown no decline, except in one study, where a decrease of approximately 10–14% was found in 42 elite players observed across 129 matches (25). The same study found a decrease in high-intensity running between halves of approximately 10%, which is slightly larger than in the present study, but in line with other previous studies (11,23,34). However, conflicting evidence exists, as other studies find no difference between halves regarding high-intensity running (16,19). Previous studies have found decreases in the number of accelerations of approximately 4–9% (23,35,38), whereas one study found no reduction in the amount of accelerations between halves (10). The discrepancy observed between studies highlights the variability in soccer, possibly because of the independent characteristics of the league, the level of the team, level of opposition, fitness level, tactical approach, and the like (4,5,8–11,25,31). Nonetheless, it is not necessarily a negative outcome itself that the distance covered decreases in the second half, as it has been shown that significant larger reductions in high-intensity running and TD covered can occur in matches heavily won vs. no difference between halves during heavy losses (11).

Also, very interestingly, central players demonstrated a large decrease in performance between halves with no difference being observed for wide players for any parameters. This is surprising, as wide players overall have higher or as high game demands and it has previously been shown that the players with the highest physical performance in the first half, usually have more difficulty maintaining their performance throughout the game (11). Looking at the specific positions, the decrease in high-intensity running and sprinting turned out significant only for central midfielders, whereas TD and the number of accelerations and decelerations decreased for both central midfielders and central defenders. For attackers, the same trends were observed, but this did not turn out significant, possibly because of the low number of observations for this position ($n = 5$). The results of the present study with a significant decrease in the performance in particular for the central players contrasts previous findings and could possibly be related to tactical alterations during the games that was won quite clearly (12,23,36).

The number of accelerations and decelerations and TD covered declined gradually during each half, when evaluating each 15-minute period, whereas no differences were present for high-intensity running or sprinting. Thus, more accelerations and decelerations and TD were covered during the first 30 minutes of the game and the first 15-minute period of the second half compared with the final half hour of the match. This pattern is similar to what was observed in a study by Akenhead et al. (1). As such, the physical performance of the players seem to decrease gradually throughout

each half, being temporarily recovered at the onset of the first half. Based on the current findings, accelerations and decelerations appear to be sensitive to fatigue during matches and thus could present reliable measures of physical match performance, as previously suggested (1). This is supported by findings in one study, where the frequency of accelerations decreased across a congested youth tournament, with no changes occurring for high-intensity running or sprinting (2). Although the exact mechanisms are not fully understood, reductions in central neural drive and increases in peripheral fatigue have previously shown to reduce rate of force development and maximal voluntary contraction strength (18,24,27,30,37), which could attenuate the ability to accelerate and decelerate during the final stages of a game. In addition, it has been suggested that when a player is tired and has a reduced acceleration output, it will take longer to reach a given speed and that the player may therefore be required to cover a greater distance at high speed (1,35). Contrasting these findings, other previous studies have not found the same reduction in acceleration capacity in comparison to high-intensity running or sprinting throughout a game (10,25,35). As such, more studies are needed with a more detailed statistical analysis than in the present study, to explore the sensitivity of accelerations and decelerations across matches and during congested periods. Contrasting the hypothesis, the match performance of the youth players of the present study was equal to the performance of the senior players. In fact, the U19 players covered even more TD and a higher number of accelerations and decelerations per minute. This is surprising because elite players have been shown to be able to produce superior performances during the Yoyo-Intermittent recovery tests, which have been shown to be good predictors of physical match performance (5). However, the findings of the present study are in line with the results obtained by another study where approximately 123–126 accelerations were recorded during 2 experimental junior soccer matches, which is considerably higher than the number performed by senior players in the present study (~81) (29). This could be related to the style of play during youth soccer matches, and the lack of experience needed for the players to pace themselves through the game. Finally, body mass, muscle strength, and explosive capability have been found to be superior in senior soccer players, potentially raising the intensity of the game because of more intense body impacts, tackles, sprints, and higher intensity accelerations and decelerations (21,22). However, based on the current evidence, youth soccer players seem to be well capable of producing similar physical match performances as senior players.

Some limitations of the present study exist, as the sample size is relative small. Moreover, this was especially the case for attackers, as only 5 observations were made at senior level for this position. Thus, it cannot be excluded that the results would have been different if a whole season was recorded. Furthermore, no statistical corrections were made for the fact that different players contributed with a varying

number of observations for each position. Meanwhile, the games at senior level were all, except one, played on home ground and won quite clearly with a superior goal difference, which could have influenced the activities during the matches as previously observed (11,19,31). Moreover, the tactical approach by this team differs from other teams, and it would be interesting for future studies to examine the differences in physical match performance between teams with different playing styles and top and bottom teams. Finally, one should be aware of the discrepancy that can occur when comparing the present results with results from other studies using different tracking systems or different velocity thresholds (13,32,39).

PRACTICAL APPLICATIONS

The main findings of the present study demonstrated that wide players performed more accelerations and decelerations than central players, with central defenders performing the lowest number of all positions. Therefore, strength and conditioning specialists should consider the higher load put on wide players and prepare them for the demands of competition by designing position-specific training drills with a higher number of accelerations and decelerations. In contrast, central defenders could instead of quantity of actions focus more on developing the quality of accelerations and decelerations through training, targeting their ability to develop power in short intense sequences during the game. In addition, youth players performed as much high-intense activities as the senior players, demonstrating that this is not a discriminating physiological ability between these levels of play and thus should not be a specific focus in the training. Finally, accelerations and decelerations seem as valid and sensitive markers of physical performance and should be monitored closely to detect fatigue during and between games, especially during congested periods.

ACKNOWLEDGMENTS

The study was conducted in cooperation with FC Midtjylland and the authors thank the club and their performance analyst Mads Bendiksen for the cooperation. No external funding was received for the study and the authors have no conflict of interest to disclose. The results of the present study do not constitute endorsement by the NSCA.

REFERENCES

1. Akenhead, R, Hayes, PR, Thompson, KG, and French, D. Diminutions of acceleration and deceleration output during professional football match play. *J Sci Med Sport* 16: 556–561, 2013.
2. Arruda, AF, Carling, C, Zanetti, V, Aoki, MS, Coutts, AJ, and Moreira, A. Effects of a very congested match schedule on body-load impacts, accelerations, and running measures in youth soccer players. *Int J Sports Physiol Perform* 10: 248–252, 2015.
3. Bangsbo, J. The physiology of soccer—with special reference to intense intermittent exercise. *Acta Physiol Scand Suppl* 619: 1–155, 1994.
4. Bangsbo, J, Iaia, FM, and Krstrup, P. Metabolic response and fatigue in soccer. *Int J Sports Physiol Perform* 2: 111–127, 2007.

5. Bangsbo, J, Iaia, FM, and Krstrup, P. The Yo-Yo intermittent recovery test: A useful tool for evaluation of physical performance in intermittent sports. *Sports Med* 38: 37–51, 2008.
6. Bendiksen, M, Pettersen, SA, Ingebrigtsen, J, Randers, MB, Brito, J, Mohr, M, Bangsbo, J, and Krstrup, P. Application of the Copenhagen soccer test in high-level women players—locomotor activities, physiological response and sprint performance. *Hum Mov Sci* 32: 1430–1442, 2013.
7. Bloomfield, J, Polman, R, and O'Donoghue, P. Physical demands of different positions in FA premier league soccer. *J Sports Sci Med* 6: 63–70, 2007.
8. Bradley, PS, Carling, C, Archer, D, Roberts, J, Dodds, A, Di Mascio, M, Paul, D, Diaz, AG, Peart, D, and Krstrup, P. The effect of playing formation on high-intensity running and technical profiles in english FA premier league soccer matches. *J Sports Sci* 29: 821–830, 2011.
9. Bradley, PS, Carling, C, Gomez Diaz, A, Hood, P, Barnes, C, Ade, J, Boddy, M, Krstrup, P, and Mohr, M. Match performance and physical capacity of players in the top three competitive standards of english professional soccer. *Hum Mov Sci* 32: 808–821, 2013.
10. Bradley, PS, Di Mascio, M, Peart, D, Olsen, P, and Sheldon, B. High-intensity activity profiles of elite soccer players at different performance levels. *J Strength Cond Res* 24: 2343–2351, 2010.
11. Bradley, PS and Noakes, TD. Match running performance fluctuations in elite soccer: Indicative of fatigue, pacing or situational influences? *J Sports Sci* 31: 1627–1638, 2013.
12. Bradley, PS, Sheldon, W, Wooster, B, Olsen, P, Boanas, P, and Krstrup, P. High-intensity running in english FA premier league soccer matches. *J Sports Sci* 27: 159–168, 2009.
13. Buchheit, M, Allen, A, Poon, TK, Modonutti, M, Gregson, W, and Di Salvo, V. Integrating different tracking systems in football: Multiple camera semi-automatic system, local position measurement and GPS technologies. *J Sports Sci* 32: 1844–1857, 2014.
14. Carling, C, Gregson, W, McCall, A, Moreira, A, Wong del, P, and Bradley, PS. Match running performance during fixture congestion in elite soccer: Research issues and future directions. *Sports Med* 45: 605–613, 2015.
15. Di Salvo, V, Baron, R, Gonzalez-Haro, C, Gormasz, C, Pigozzi, F, and Bachl, N. Sprinting analysis of elite soccer players during European champions league and UEFA cup matches. *J Sports Sci* 28: 1489–1494, 2010.
16. Di Salvo, V, Baron, R, Tschann, H, Calderon Montero, FJ, Bachl, N, and Pigozzi, F. Performance characteristics according to playing position in elite soccer. *Int J Sports Med* 28: 222–227, 2007.
17. Di Salvo, V, Pigozzi, F, Gonzalez-Haro, C, Laughlin, MS, and De Witt, JK. Match performance comparison in top english soccer leagues. *Int J Sports Med* 34: 526–532, 2013.
18. Draganidis, D, Chatzinikolaou, A, Avloniti, A, Barbero-Alvarez, JC, Mohr, M, Malliou, P, Gourgoulis, V, Deli, CK, Douroudos, II, Margonis, K, Gioxidis, A, Flouris, AD, Jamurtas, AZ, Koutedakis, Y, and Fatouros, IG. Recovery kinetics of knee flexor and extensor strength after a football match. *PLoS One* 10: e0128072, 2015.
19. Dupont, G, Nedelec, M, McCall, A, McCormack, D, Berthoin, S, and Wisloff, U. Effect of 2 soccer matches in a week on physical performance and injury rate. *Am J Sports Med* 38: 1752–1758, 2010.
20. Gaudino, P, Iaia, FM, Alberti, G, Strudwick, AJ, Atkinson, G, and Gregson, W. Monitoring training in elite soccer players: Systematic bias between running speed and metabolic power data. *Int J Sports Med* 34: 963–968, 2013.
21. Haugen, T, Tonnessen, E, Hisdal, J, and Seiler, S. The role and development of sprinting speed in soccer. *Int J Sports Physiol Perform* 9: 432–441, 2014.
22. Hoshikawa, Y, Iida, T, Muramatsu, M, Nakajima, Y, Fukunaga, T, and Kanehisa, H. Differences in thigh muscularity and dynamic torque between junior and senior soccer players. *J Sports Sci* 27: 129–138, 2009.
23. Ingebrigtsen, J, Dalen, T, Hjelde, GH, Drust, B, and Wisloff, U. Acceleration and sprint profiles of a professional elite football team in match play. *Eur J Sport Sci* 15: 101–110, 2015.
24. Krstrup, P, Ortenblad, N, Nielsen, J, Nybo, L, Gunnarsson, TP, Iaia, FM, Madsen, K, Stephens, F, Greenhaff, P, and Bangsbo, J. Maximal voluntary contraction force, SR function and glycogen resynthesis during the first 72 h after a high-level competitive soccer game. *Eur J Appl Physiol* 111: 2987–2995, 2011.
25. Mohr, M, Krstrup, P, and Bangsbo, J. Match performance of high-standard soccer players with special reference to development of fatigue. *J Sports Sci* 21: 519–528, 2003.
26. Nedelec, M, McCall, A, Carling, C, Legall, F, Berthoin, S, and Dupont, G. The influence of soccer playing actions on the recovery kinetics after a soccer match. *J Strength Cond Res* 28: 1517–1523, 2014.
27. Nybo, L, Girard, O, Mohr, M, Knez, W, Voss, S, and Racinais, S. Markers of muscle damage and performance recovery after exercise in the heat. *Med Sci Sports Exerc* 45: 860–868, 2013.
28. Osgnach, C, Poser, S, Bernardini, R, Rinaldo, R, and di Prampero, PE. Energy cost and metabolic power in elite soccer: A new match analysis approach. *Med Sci Sports Exerc* 42: 170–178, 2010.
29. Pettersen, SA, Krstrup, P, Bendiksen, M, Randers, MB, Brito, J, Bangsbo, J, Jin, Y, and Mohr, M. Caffeine supplementation does not affect match activities and fatigue resistance during match play in young football players. *J Sports Sci* 32: 1958–1965, 2014.
30. Rampinini, E, Bosio, A, Ferraresi, I, Petruolo, A, Morelli, A, and Sassi, A. Match-related fatigue in soccer players. *Med Sci Sports Exerc* 43: 2161–2170, 2011.
31. Rampinini, E, Coutts, AJ, Castagna, C, Sassi, R, and Impellizzeri, FM. Variation in top level soccer match performance. *Int J Sports Med* 28: 1018–1024, 2007.
32. Randers, MB, Mujika, I, Hewitt, A, Santisteban, J, Bischoff, R, Solano, R, Zubillaga, A, Peltola, E, Krstrup, P, and Mohr, M. Application of four different football match analysis systems: A comparative study. *J Sports Sci* 28: 171–182, 2010.
33. Reilly, T. Energetics of high-intensity exercise (soccer) with particular reference to fatigue. *J Sports Sci* 15: 257–263, 1997.
34. Rey, E, Lago-Peñas, C, Lago-Ballesteros, J, Casais, L, and Dellal, A. The effect of a congested fixture period on the activity of elite soccer players. *Biol Sport* 27: 181–185, 2010.
35. Russell, M, Sparkes, W, Northeast, J, Cook, CJ, Love, TD, Bracken, RM, and Kilduff, LP. Changes in acceleration and deceleration capacity throughout professional soccer match-play. *J Strength Cond Res* 30: 2839–2844, 2016.
36. Terje, D, Jorgen, I, Gertjan, E, Geir Havard, H, and Ulrik, W. Player load, acceleration, and deceleration during 45 competitive matches of elite soccer. *J Strength Cond Res* 30: 351–359, 2016.
37. Thorlund, JB, Aagaard, P, and Madsen, K. Rapid muscle force capacity changes after soccer match play. *Int J Sports Med* 30: 273–278, 2009.
38. Varley, MC and Aughey, RJ. Acceleration profiles in elite Australian soccer. *Int J Sports Med* 34: 34–39, 2013.
39. Varley, MC, Fairweather, IH, and Aughey, RJ. Validity and reliability of GPS for measuring instantaneous velocity during acceleration, deceleration, and constant motion. *J Sports Sci* 30: 121–127, 2012.
40. Wehbe, GM, Hartwig, TB, and Duncan, CS. Movement analysis of Australian national league soccer players using global positioning system technology. *J Strength Cond Res* 28: 834–842, 2014.